FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

AARHUS
UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT

TLDR

Two hours finding out what actuators feel like, and leaving able to drive one from your own code.

Get a board and run task 01. Something spins.

Change two numbers, upload again. That loop is the workshop.

Pick up the actuators you are not driving.

Work through as many of the five tasks as you get to.

Build one signal someone else can read without looking.

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew the lab has 117 coin motors

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the tables at the front

- Go and feel the ones your task does not use

Finish by building one signal someone else can read

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

```
0:00     10 min    Two motors, same message. Which felt urgent?

0:10     15 min    Boards out, task 01 running for everyone

0:25     60 min    Tasks 02 to 05, at your own pace

1:25     10 min    Round-the-room: what surprised you

1:35     20 min    Blind test on task 05

1:55      5 min    Pack down
```

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

01	**Make something spin**	**coin motor**
02	**Make something tap**	**solenoid**
03	**Make something warm**	**Peltier tile**
04	**Make it notice you**	**a wire and foil**
05	**Make it answer back**	**both together**
06	**Use what you have**	**sensors.chomskylab.dk**

1 to 3 output, 4 input, 5 both. Nobody finishes all six.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire. All of them:
github.com/labtools-au/multimodal-actuator-workshop

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one

PIN MAP

01	spin	D9 PWM	1k to PN2222 base, motor on collector
02	tap	D6	MOSFET gate, own supply, grounds joined
03	warm	D3 + D5 PWM	H-bridge low and high, bench supply
04	sense	A0	bare wire to foil. That is all of it
05	answer	A0 + D9	the pad, plus any actuator
07	stepper	D8 D9 D10 D11	ULN2003 IN1 to IN4

Only D3, D5, D6, D9, D10, D11 do PWM. Stepper pins go into the constructor OUT of order: 8, 10, 9, 11.

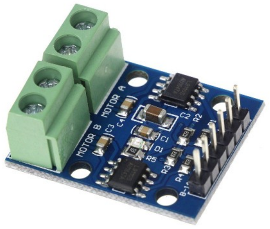
WHAT YOU GET IN THE BASE KIT

One set per pair, collected before you start task 01. Below: the three parts people always come back for.

Board, breadboard, jumper wires, resistors: on the tables.

Not in the drawers, so take what you need.

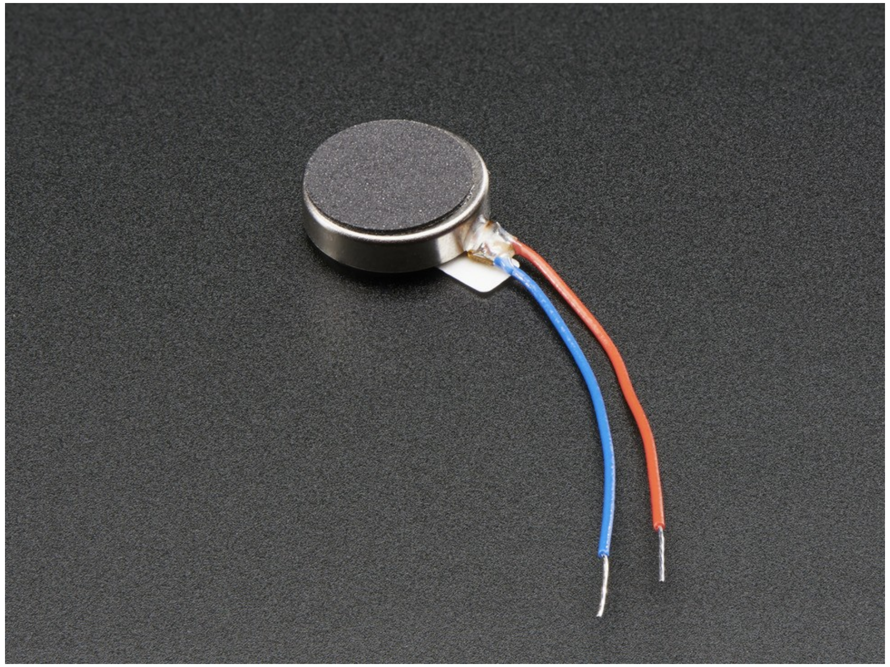
Breadboard supply, 8A. Anything past an LED.



L9110S H-bridge, 2C. Reversing, and the Peltier.

PAM8403 amp, 11F. A transducer needs one.

VIBRATING MINI MOTOR DISC



An off-centre weight on a motor shaft, sealed in a disc. The same part as in your phone.

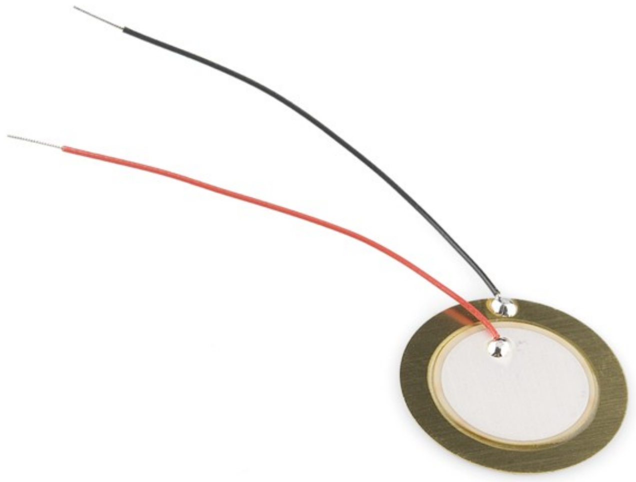
Feels like: a buzz you cannot make gentle and fast at once.

USED FOR

Phone notifications, controller rumble, anything worn under clothing.

`1E | 117 | 2.5-3.8V rated | 100 mA at 5V | PWM`

PIEZO ELEMENT



A crystal that flexes when you put current across it. Near-instant, almost no power, but shallow.

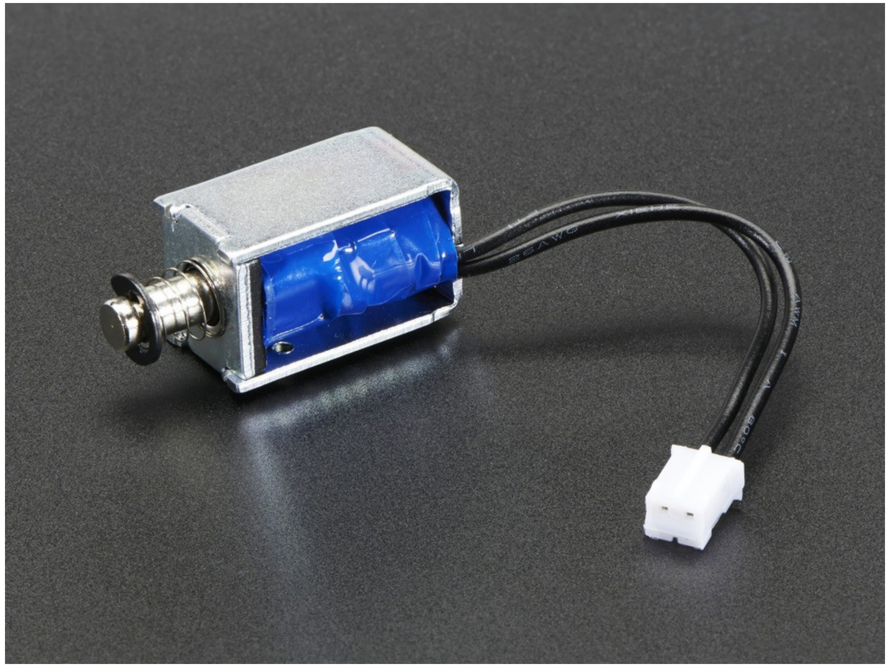
Feels like: a tick, not a thump.

USED FOR

Clicks and confirmations. Anything where lag would feel broken.

`Drawer 6F  |  69 in stock  |  any digital pin`

MINI PUSH-PULL SOLENOID



A coil that yanks a metal rod when you energise it. No half a tap: it fires or it does not.

Feels like: a person tapping you, not a machine signalling.

USED FOR

Navigation cues on the body, braille cells, alerts in noise.

`3E | 24 | 5V, 1.1 A, 4.5 ohm | 3mm throw, 80g`

CAPACITIVE PAD

PHOTO
Capacitive pad

A wire taped behind foil, card or fabric. No sensor and no breakout board, which is why the box on the left is empty.

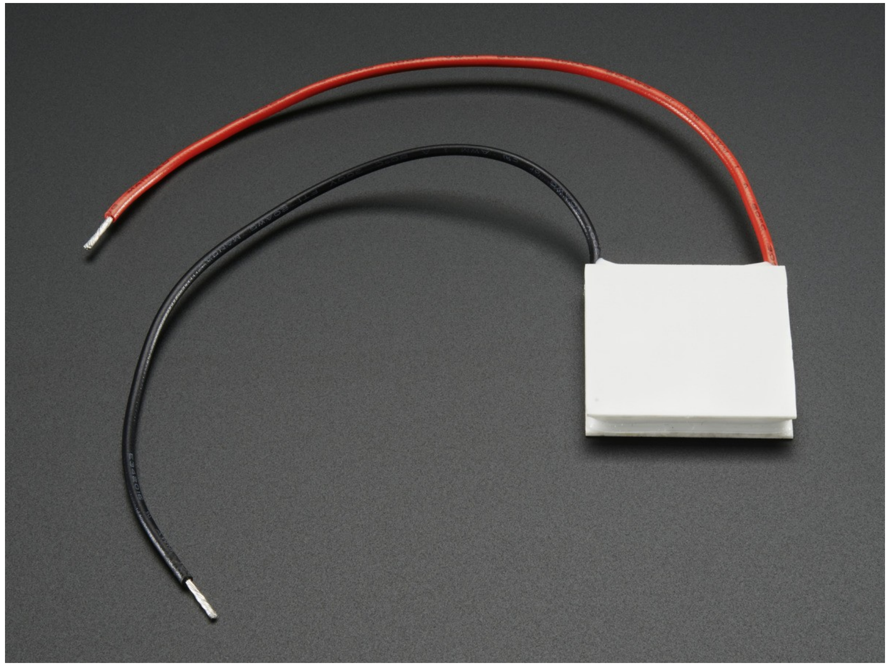
Feels like: nothing. That is the point. The surface stays plain.

USED FOR

Invisible controls in wood or fabric, waterproof panels.

`No drawer, no part number, no cost. Any analogue pin`

PELTIER TILE



Heats one face and cools the other. Flip the current and it reverses.

Feels like: slow. Several seconds before you are sure which way.

USED FOR

Slow ambient state. Never an alert.

`Drawer 4C | 25 small, 35 big | needs an H bridge`

SURFACE TRANSDUCER

Turns any surface into a speaker. Drive it so one frequency is felt and another is heard, from the same part.

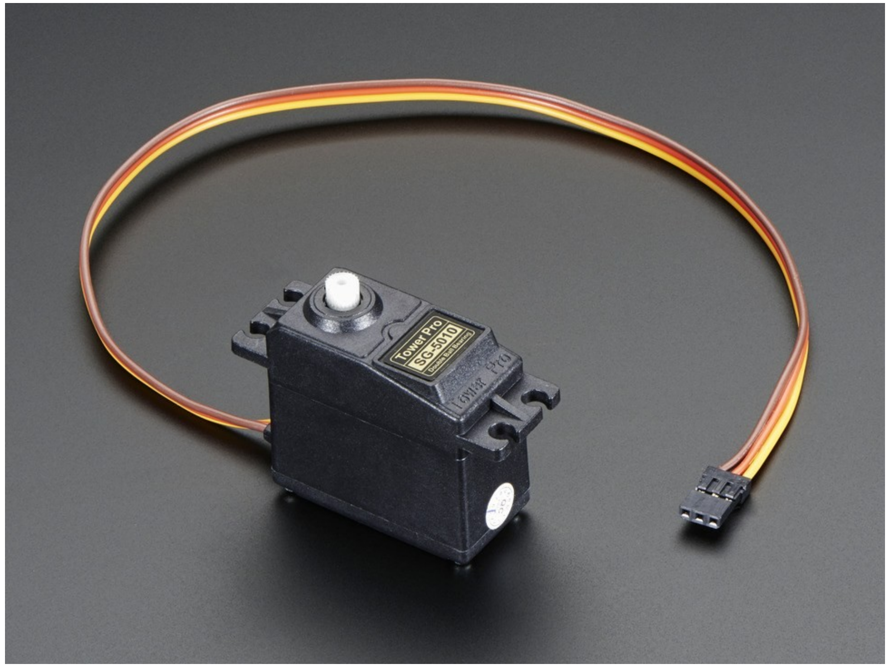
Feels like: more certain, rather than louder.

USED FOR

Noisy or bright environments, and accessibility.

`Drawer 7F  |  12 large, 13 medium, 23 bone  |  needs an amp`

SERVO



A motor that holds a position instead of spinning freely. Tell it an angle and it goes there.

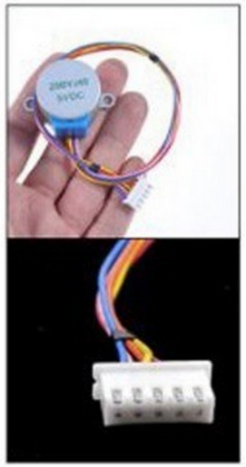
Feels like: something deliberate moving, with force behind it.

USED FOR

Anything that points, pushes, opens or resists a hand.

`Drawer 2F  |  80 in stock  |  Servo.h  |  needs its own 5V`

STEPPER MOTOR



Moves in fixed steps rather than continuously, so you always know where it is without a sensor.

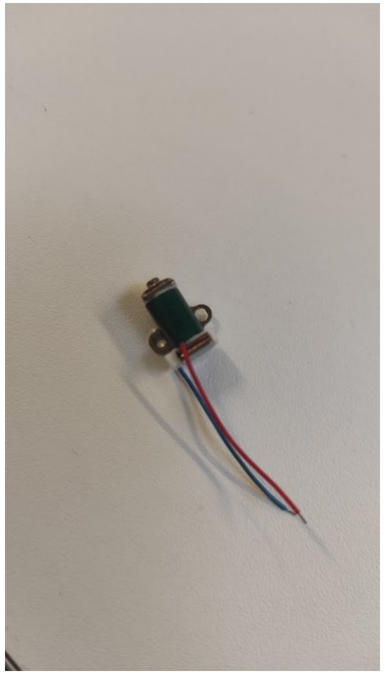
Feels like: precise, and audibly clicky.

USED FOR

Slow accurate motion. Dials, sliders, anything positioned.

`2A | 44 | ULN2003 driver | 2048 steps/rev`

ELECTROMAGNET



Grabs and releases ferrous metal on command. No moving parts of its own.

Feels like: a grip that appears and vanishes.

USED FOR

Latches, holds, and anything that should let go on cue.

`Drawer 3D | 17 mini, 11 standard | needs a MOSFET`

FREESTYLE, WITHIN THREE LIMITS

You do not have to do the tasks in order, or finish them. Playing properly with two beats rushing all five.

If you want to drive something not in front of you, go get it.

The three things that are not negotiable

• Solenoid and Peltier get their own supply, never USB

• The Peltier heatsink goes on before it is switched on

• No motor straight from a pin. 40 mA limit, it wants 75

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

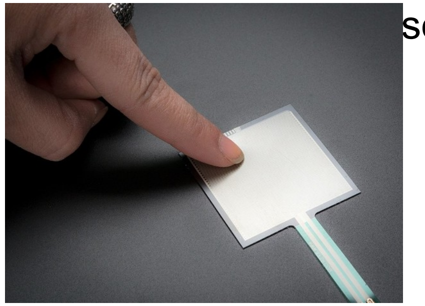
INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- GSR, drawer 0C5. Exactly one in the lab, so ask first

so



Force (FSR), 4A. 46 tiny, 33 square. Calibrate each one.

Flex, 4B. Only 8 left, so share them.

Pulse, 9D. 9 in stock. Steady at rest, useless once moving.

GUSTAV SIMONSEN
TEACHING ASSISTANT

YOU ALREADY WRITE CODE. WHY COME?

Fair question. Three honest answers.

You cannot Google what something feels like

- No datasheet says a knock reads as a person, a buzz as a machine

The parts are free and already here

- Borrowing beats ordering. Nothing to buy, nothing to wait for

Your devices hide sensors you may not be able to reach

- Find out today which open up, and which are locked

REVERSE ENGINEER WHAT YOU ALREADY OWN

The trackers in your pocket and on your wrist. What opens up, and what does not.

BLE heart rate strap: readable from a web page

• Web Bluetooth, standard GATT. Chrome and Edge only

Most Garmin watches: the same, with no app

• Turn on Broadcast Heart Rate and it acts as a strap

Apple Watch: nothing live, whatever you write

• HealthKit needs a watchOS app in Swift

Garmin history: Connect IQ, or the API afterwards

• Good for analysis. Useless for reacting in real time

WHAT A PHONE WILL AND WILL NOT GIVE YOU

Open sensors.chomskylab.dk on your own phone right now.

Works in a web page

- Accelerometer and gyroscope, microphone, camera
- Pointer pressure, though a trackpad reports only 0 or 0.5

Does not, whatever the tutorial says

- Phone vibration on iOS. WebKit has never shipped it, at any version, and Chrome on iOS is Safari underneath

So "the phone buzzes" works on no iPhone in this room.

- Check the feature, not the tutorial, before it is in a plan.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses, and how strong they are

Blind test: they name all three

- Note which two got confused, and why

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,

you built a visual interface with a motor glued to it.

BEFORE YOU CALL IT DONE

Hand your device to another group with the screen turned away.

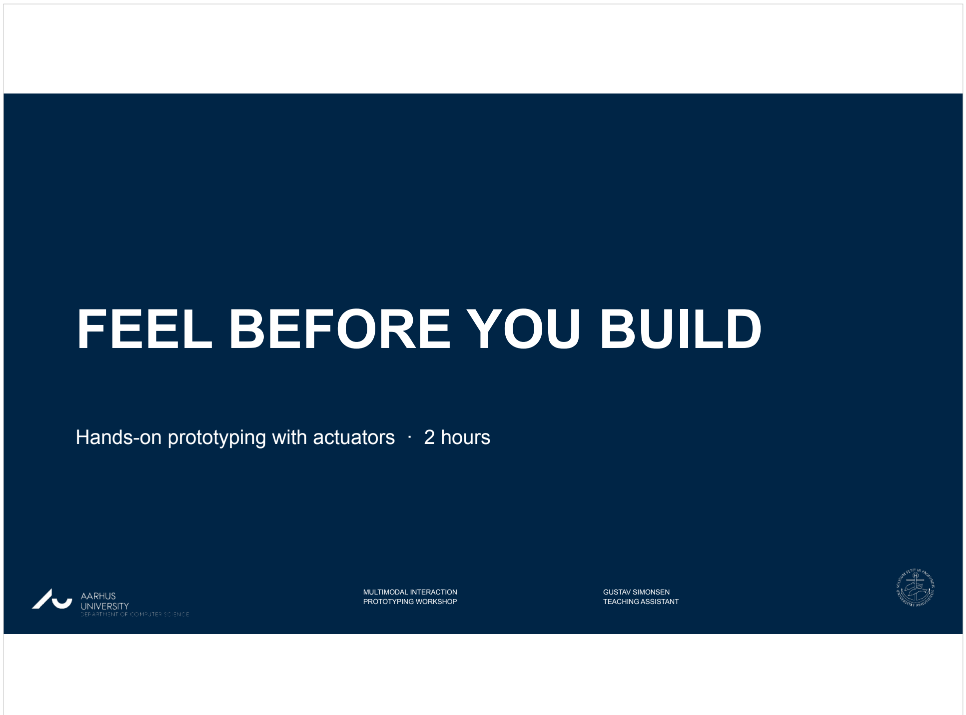
- They can tell all three messages apart
- It does not fire continuously when held
- You know which two got confused, and why

What you actually leave with

- What things feel like, how to drive one, and which to pick
- That last one is the expensive mistake this prevents

GO AND TOUCH THINGS

github.com/labtools-au/multimodal-actuator-workshop



0:00. Start here, and do not read the slide.

Open with the demo, not with talking. Have the coin motor and the solenoid both wired and running before anyone sits down.

Buzz a phone-style notification on the motor. Then send the same message on the solenoid. Ask which felt more urgent. Let them answer. Nobody needs theory for that, and it is the argument for the whole session.

TLDR

Two hours finding out what actuators feel like, and leaving able to drive one from your own code.

Get a board and run task 01. Something spins.

Change two numbers, upload again. That loop is the workshop.

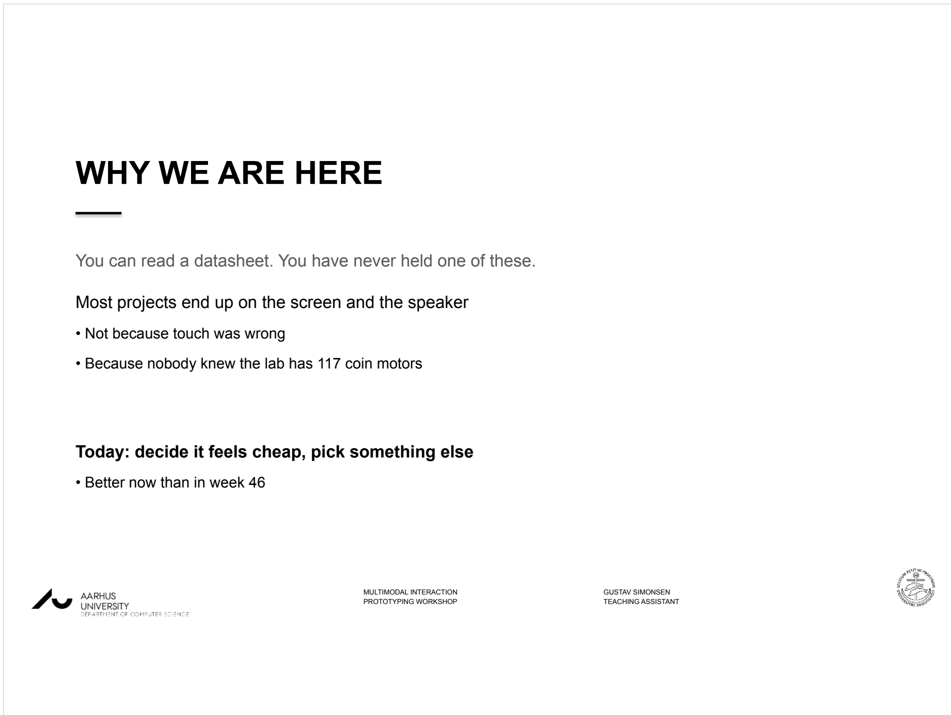
Pick up the actuators you are not driving.

Work through as many of the five tasks as you get to.

Build one signal someone else can read without looking.

Fifteen seconds. It is a map, not a talk.

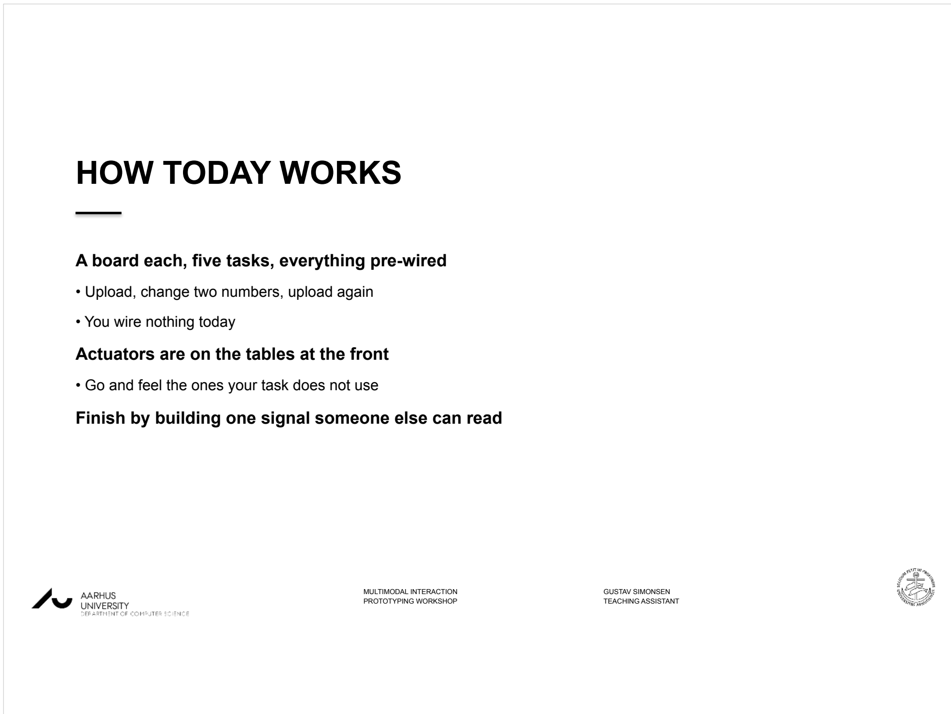
Useful because half the room arrives expecting a lecture and needs to be told immediately that this is not one.



Say the full version out loud, the slide only carries the skeleton:

"You can read a datasheet. You have never held one of these. Today you put a vibration motor against your own wrist, decide it feels cheap, and pick something else. Better now than in week 46 with the demo three days out."

Every year projects end up on the screen and the speaker, and touch gets designed on paper then dropped. Two minutes, then move.



Set the shape clearly, because it is not a lecture and they will wait to be lectured at.

A board each. Five tasks. Everything pre-wired. They upload, change numbers, upload again.

The actuators they are NOT driving sit on tables around the room. Tell them explicitly to go and pick those up between tasks. The comparison still happens, just on their own initiative rather than on a timer.

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Boards out, task 01 running for everyone
0:25	60 min	Tasks 02 to 05, at your own pace
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down

Show it, do not narrate it. Fifteen seconds.

The one hard checkpoint is 0:25. Everyone should have task 01 running by then.
If someone is still fighting drivers at 0:30, pair them with a group that is
working rather than debugging it alone.

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

01	**Make something spin**	**coin motor**
02	**Make something tap**	**solenoid**
03	**Make something warm**	**Peltier tile**
04	**Make it notice you**	**a wire and foil**
05	**Make it answer back**	**both together**
06	**Use what you have**	**sensors.chomskylab.dk**

1 to 3 output, 4 input, 5 both. Nobody finishes all six.

AARHUS UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

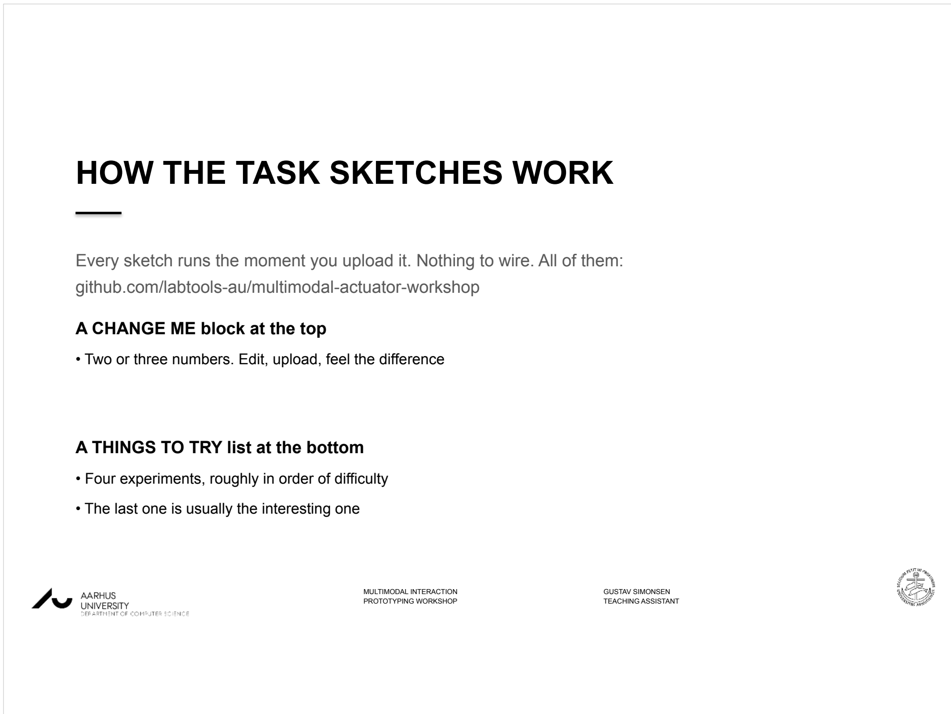
GUSTAV SIMONSEN
TEACHING ASSISTANT

Ninety seconds on this slide, no more.

Name the arc, it is the point: 1 to 3 are output, 4 is input, 5 is both. By task 5 they have built a complete interaction loop, which is the shape of most project work.

Say clearly that nobody is expected to finish all five. Getting 01 working and then properly playing with 02 beats rushing all of them.

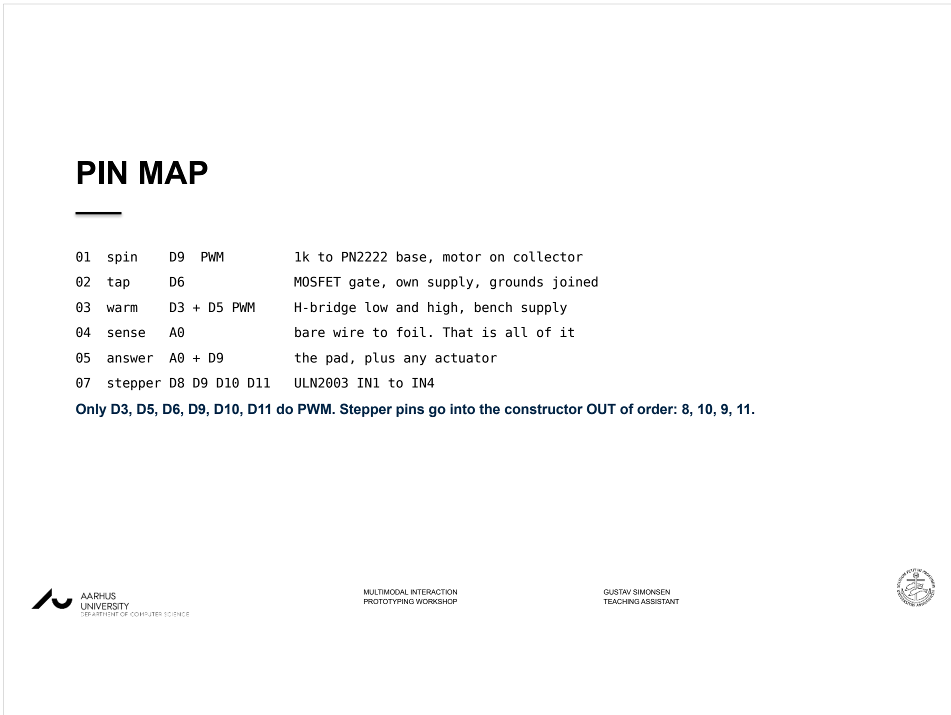
Task 05's last experiment IS the closing brief, so the work carries over.



Say the format once here rather than repeating it per task.

CHANGE ME at the top: two or three numbers, already set to something that
works. THINGS TO TRY at the bottom: four experiments, easy to hard, and the
last one usually asks for a rewrite rather than a tweak.

Upload problems, in the order they will hit you:
  * wrong board or port under Tools
  * ADCTouch not installed (tasks 04 and 05)
  * solenoid task: board resets when it fires means its supply is off



Reference, not a talking point. Ten seconds, or skip it entirely if the stations are already built.

It earns its place two ways: you use it while setting the room up, and it is the slide to jump back to when someone asks "which pin was the solenoid".

The bottom line is the one worth reading aloud if anyone touches the stepper: the pins go into the constructor OUT of order, 8, 10, 9, 11. Wired in numeric order it buzzes instead of turning and looks like a dead motor.

WHAT YOU GET IN THE BASE KIT

One set per pair, collected before you start task 01. Below: the three parts people always come back for.

Board, breadboard, jumper wires, resistors: on the tables.

Not in the drawers, so take what you need.

Breadboard supply, 8A. Anything past an LED.

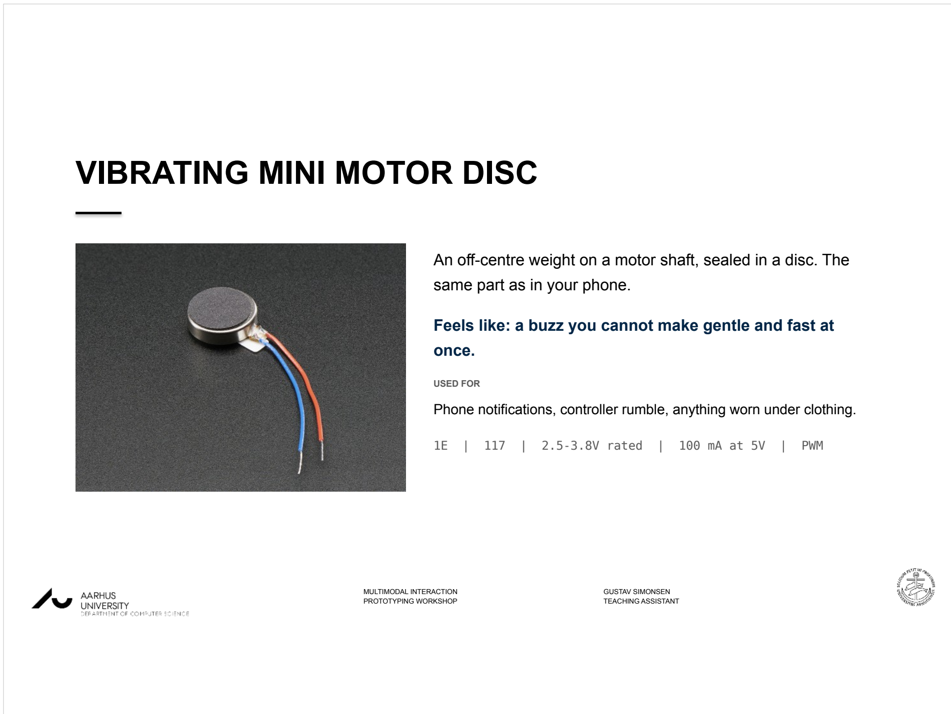
L9110S H-bridge, 2C. Reversing, and the Peltier.

PAM8403 amp, 11F. A transducer needs one.

Hand this out before anything else, or you will spend the session fetching parts one at a time.

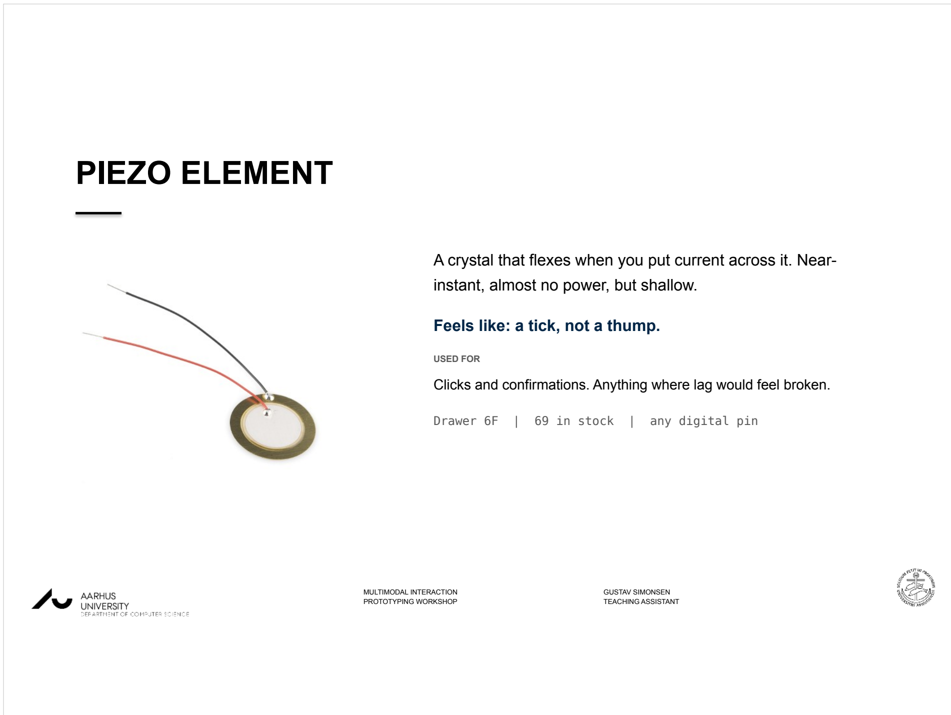
One set per pair. The three drawer numbers are the ones people actually come back and ask for: 8A breadboard supply, 2C H-bridge, 11F amplifier.

Jumper wires, breadboards and resistors are not tracked in the component database, so make sure there are enough on the tables before students arrive.



The one everybody already owns without knowing it. Every phone has one.

The point worth making: speed and strength are welded together. That single constraint is why the other five exist.



The escape from the coin motor's compromise. A crystal that flexes when you put current across it: near-instant, almost no power, but shallow.

A tick, not a thump. Point anyone here who needs something to feel responsive rather than just present. 69 in the drawer, so nobody has to share.

MINI PUSH-PULL SOLENOID

A coil that yanks a metal rod when you energise it. No half a tap: it fires or it does not.

Feels like: a person tapping you, not a machine signalling.

USED FOR

Navigation cues on the body, braille cells, alerts in noise.

`3E | 24 | 5V, 1.1 A, 4.5 ohm | 3mm throw, 80g`

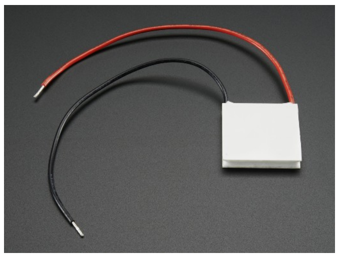
Pass this one around if you can. It is the one people remember, because a knock reads as a person rather than a machine.

Warn them now that it needs its own supply, so it does not come as a surprise at task 02.

Hold up a wire and a piece of foil. That is genuinely the whole sensor.

This is the slide that changes what people think is possible, because it costs nothing and hides completely inside a prototype.

PELTIER TILE



Heats one face and cools the other. Flip the current and it reverses.

Feels like: slow. Several seconds before you are sure which way.

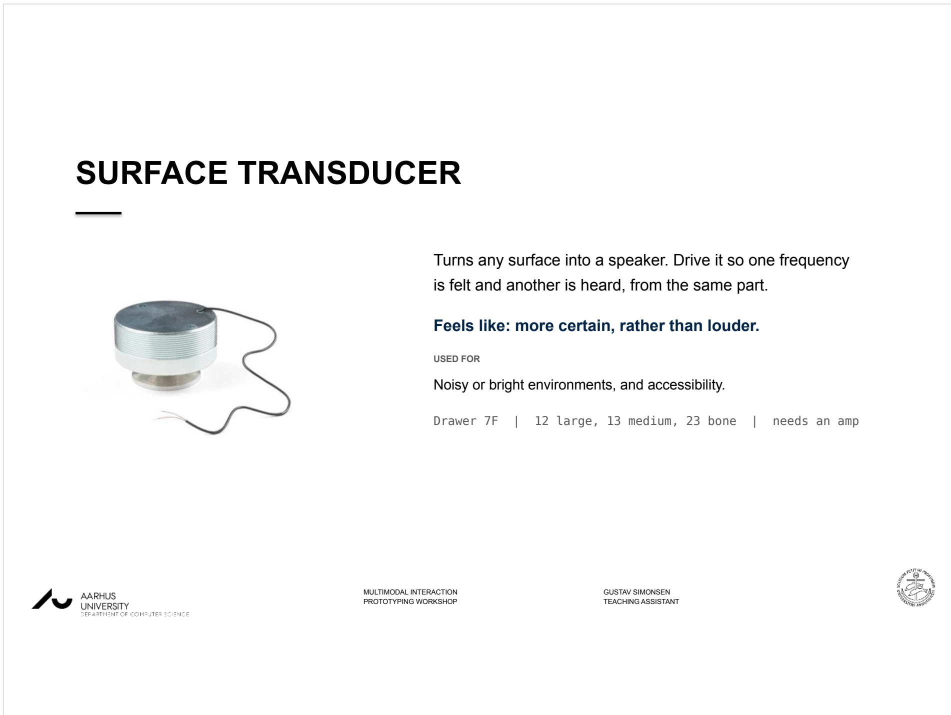
USED FOR

Slow ambient state. Never an alert.

`Drawer 4C | 25 small, 35 big | needs an H bridge`

Say the honest limitation out loud: it is slow, and hot alternating with cold just reads as lukewarm.

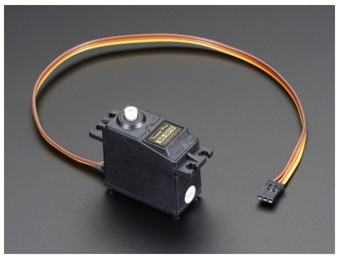
Good for slow ambient state. Wrong for alerts. Saying so here saves a project from finding out in week 46.



The redundancy argument. Same driver, one frequency you feel and one you hear.

Together is not louder, it is more certain. Cheapest reliability a project can buy, and it is also the accessibility answer.

SERVO



A motor that holds a position instead of spinning freely. Tell it an angle and it goes there.

Feels like: something deliberate moving, with force behind it.

USED FOR

Anything that points, pushes, opens or resists a hand.

`Drawer 2F | 80 in stock | Servo.h | needs its own 5V`

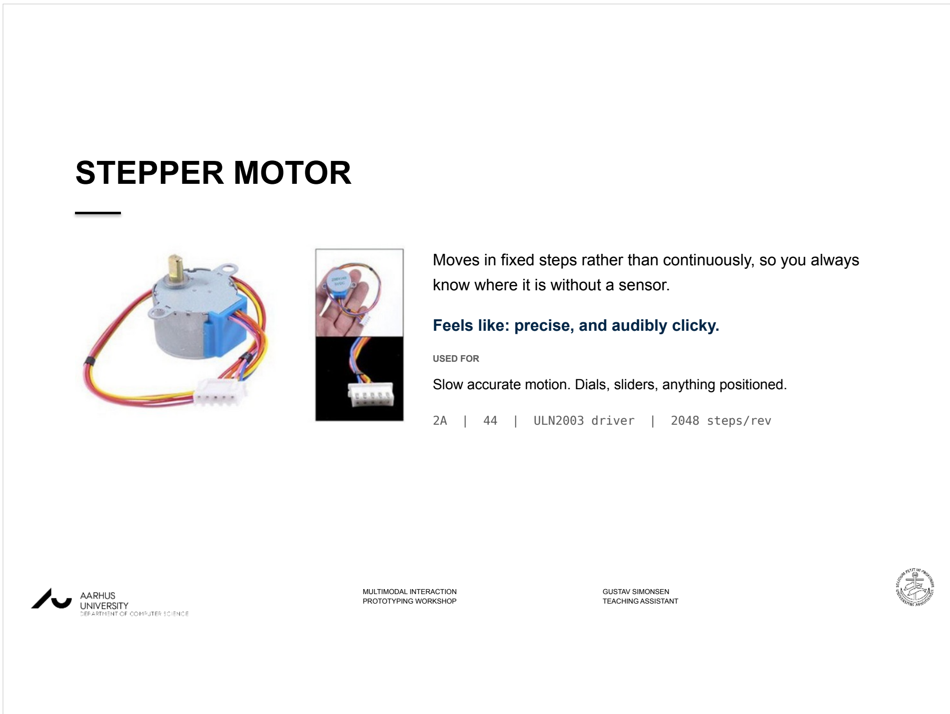
AARHUS UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT

Worth showing even though it is not in a task. Most projects that need something to physically move end up here rather than on a vibration motor.

80 in stock, so nobody has to share.

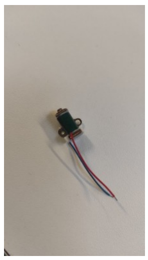


The precision option. Moves in known steps, so you can position it without any feedback sensor at all. 2048 steps per revolution.

Needs a ULN2003 driver board, drawer 2A, 27 of them.

If anyone tries one: the Stepper constructor does NOT take the pins in numeric order. Wire IN1 to IN4 on pins 8 to 11, then write Stepper(2048, 8, 10, 9, 11).
Get it wrong and it buzzes and jitters instead of turning, which looks like a dead motor. bench_tests/07_stepper has it right, with the note.

ELECTROMAGNET



Grabs and releases ferrous metal on command. No moving parts of its own.

Feels like: a grip that appears and vanishes.

USED FOR

Latches, holds, and anything that should let go on cue.

`Drawer 3D | 17 mini, 11 standard | needs a MOSFET`

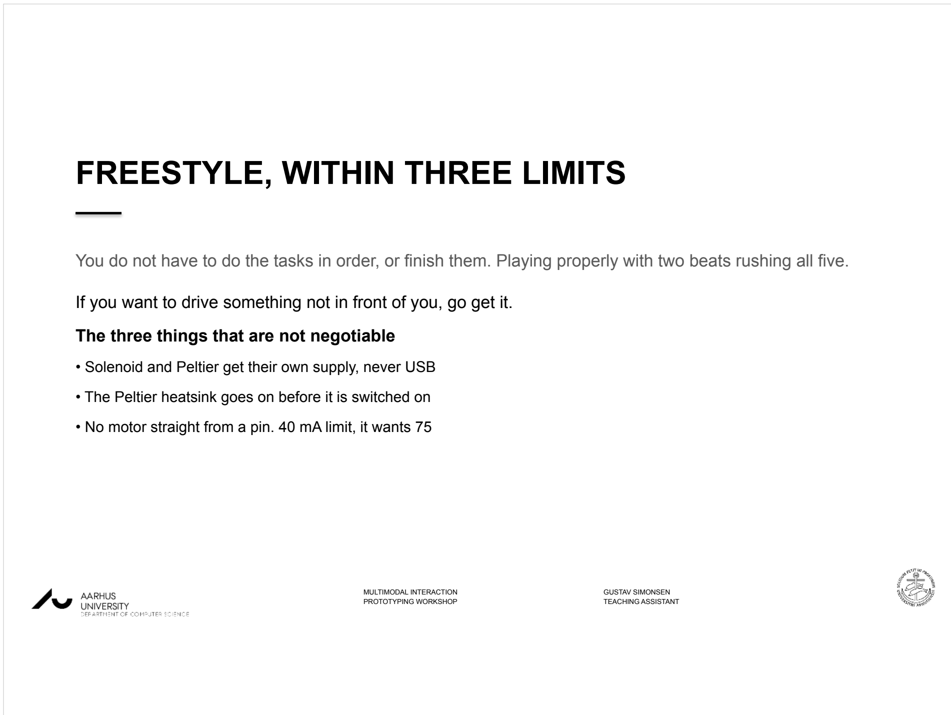
AARHUS UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT

The one people forget exists. It grabs and lets go, with nothing moving.

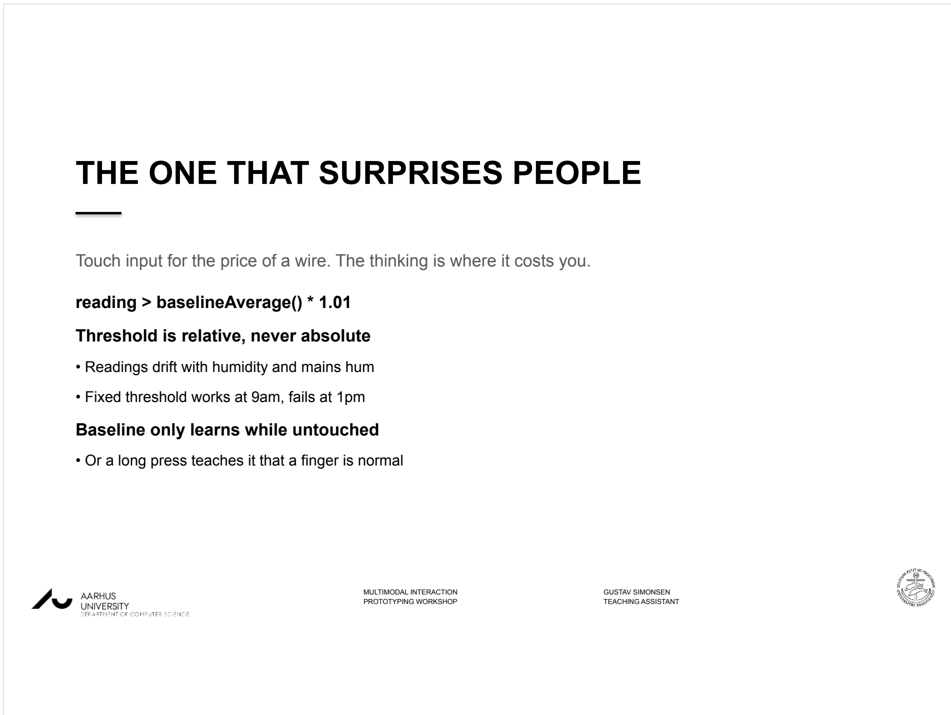
Good answer for anything that should hold and then release: a latch, a door, a
thing that falls when the event happens.



Two messages on one slide, and they pull in opposite directions on purpose. Say the permission warmly, then change tone for the limits.

Permission first: some students will not believe they are allowed to skip ahead or stop early unless told. The failure mode is a pair rushing all five badly instead of doing two properly.

Then the three hard limits, and here you should sound firm. Say the numbers out loud: a pin gives 40 mA, the motor wants about 75. It is not a rule they have to take on faith.



Slow down here. This is the single idea most worth stealing from today, and
it is worth interrupting the room for once most people have reached task 04.

Two things students get wrong:

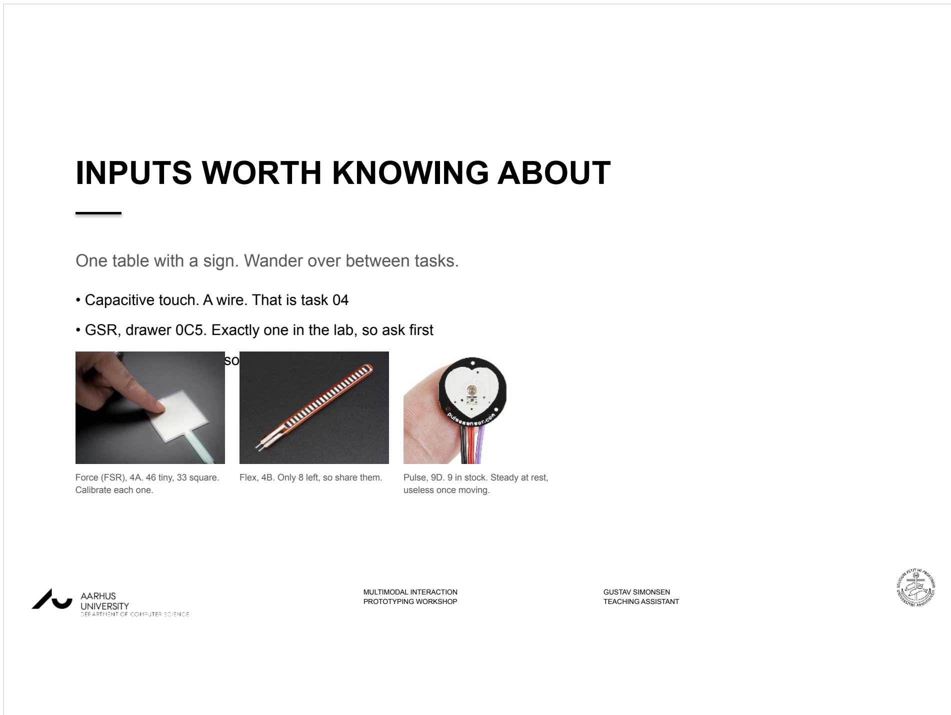
1. A fixed threshold. Works in the morning, fails after lunch, because the
   reading drifts with humidity and mains hum.

2. Feeding the baseline buffer always. A long press then teaches the baseline
   that a finger is normal, and the touch silently stops working. The buffer
   must only learn while untouched.

1:25. Pull the room back together.

One sentence per group: what surprised you. No slides, no laptops. Ten minutes.

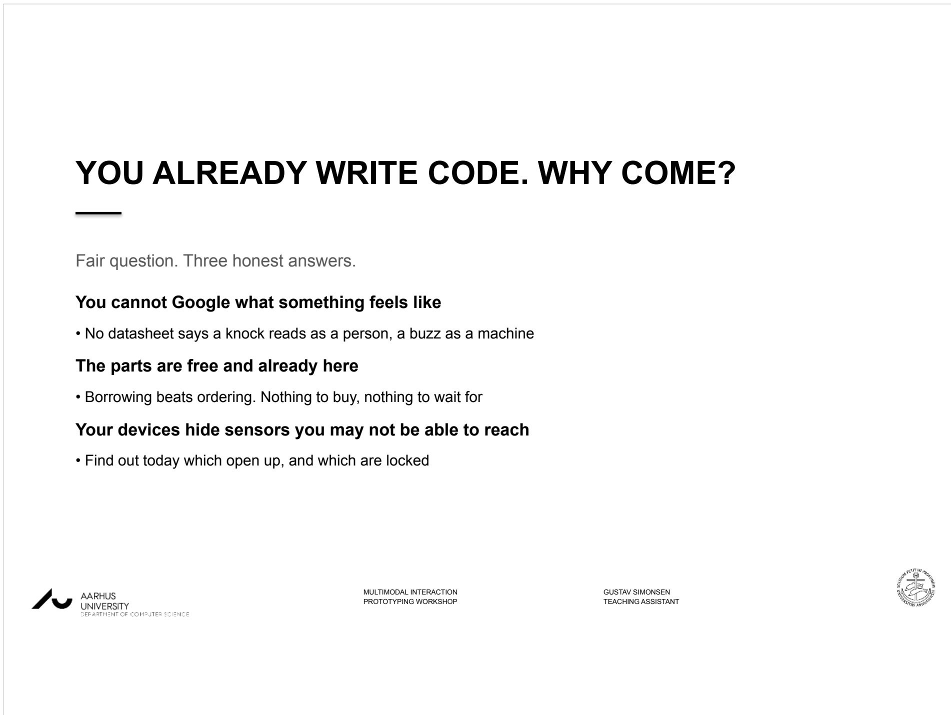
Expect someone to report the Peltier delay. Let that land, it is far more persuasive from a peer than from you.



Mention, do not teach. These sit on one table with a sign.

The honest warnings matter more than the specs: PPG is useless once the hand moves, GSR drifts so only relative change works, FSRs need per-pad calibration.

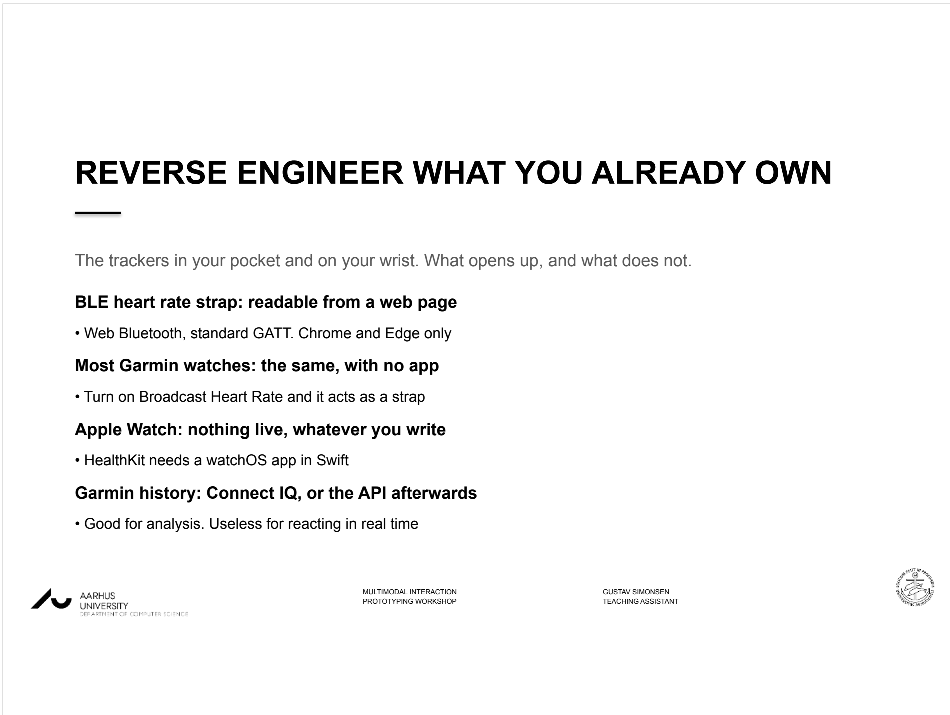
Push the phone option hard. For a lot of projects it beats everything else on the table and needs no soldering at all.



The slide that earns the room's attention, so do not rush it.

This audience can write the code already, and they know it. Saying so out loud buys credibility for the rest. The argument is not "you need to learn to program a motor", it is "you cannot Google what something feels like".

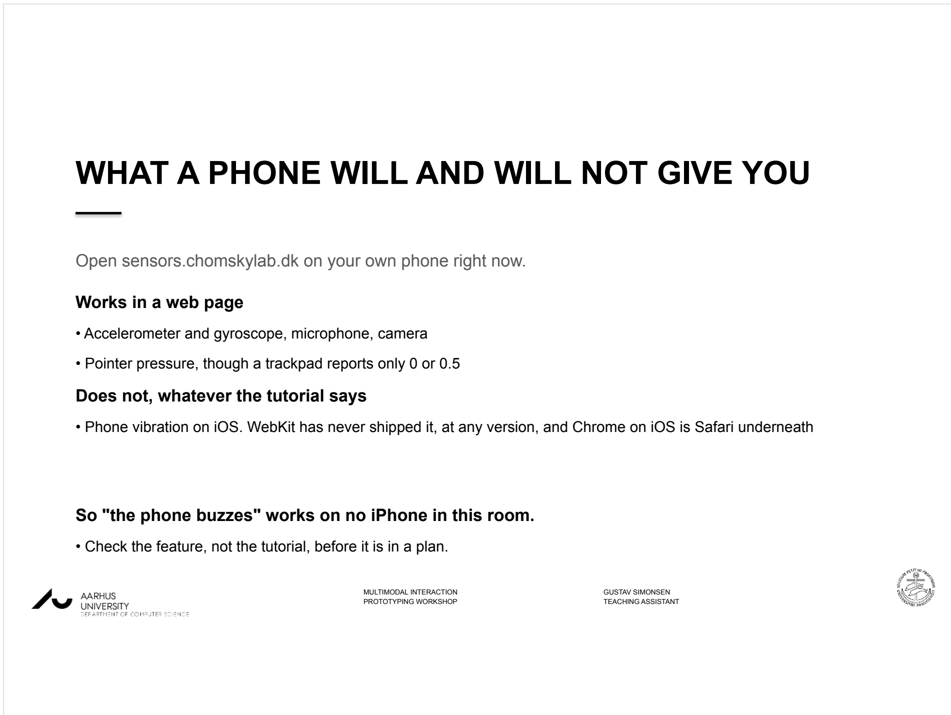
The second point is practical and lands with everyone: the parts are free and already in the building.



The slide the tinkerers will care most about, and the one where being precise matters, because the internet is full of wrong answers.

The useful surprise: most Garmin watches broadcast heart rate over BLE natively. Turn on Broadcast Heart Rate, or start a Virtual Run, and the watch appears as an ordinary heart rate strap that Web Bluetooth can read from a web page. No Connect IQ app, no SDK, no API key.

Apple Watch will not. HealthKit needs a companion watchOS app in Swift, so



Have them open sensors.chomskylab.dk on their own phones while you talk.
The split between what works and what does not lands far harder on their own device than on a slide.

The bottom half is the single most useful fact here, and the one people get wrong because a blog post said otherwise.

Say it plainly: iPhones do not vibrate from a web page. Not with a polyfill, not with a library, not in Chrome for iOS, which is Safari underneath.

The general lesson is bigger than vibration:

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

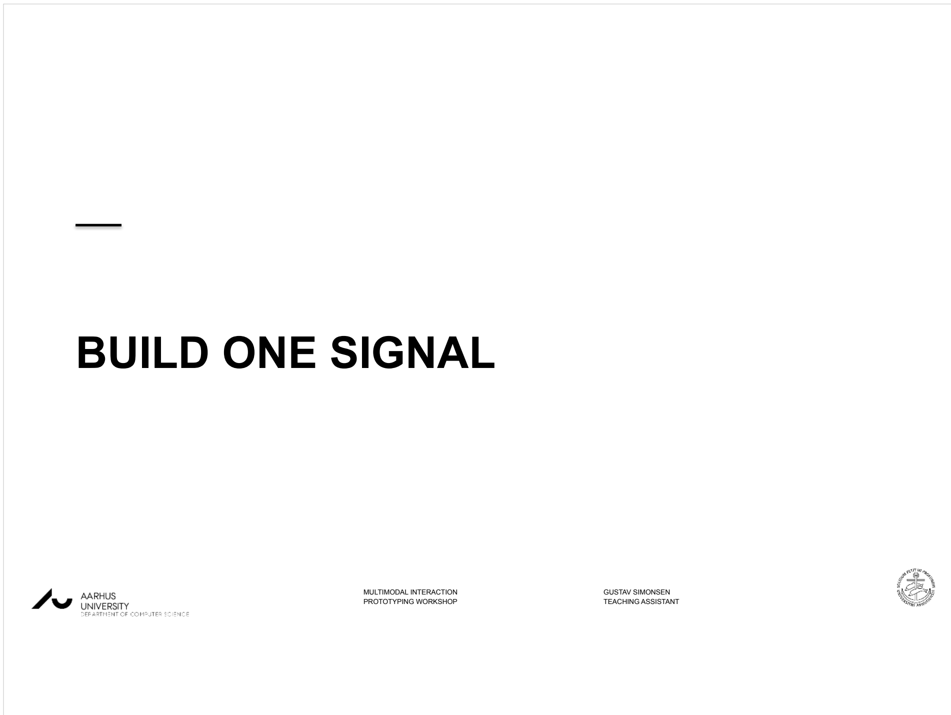
- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

Five minutes, and put the actual repos on screen if you can.

The argument: this was built by students at your stage, not by a lab with a budget. The gap between "we should use touch" and "it moves when you touch it" is smaller than it looks.

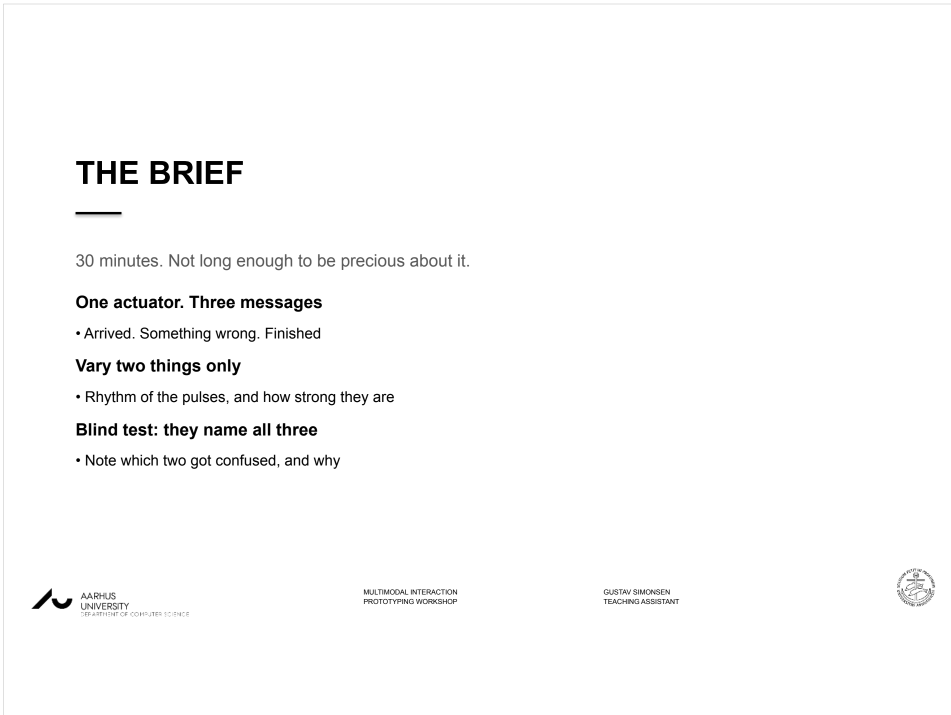
The moving screen has a browser control panel over the serial port, which is what you will wish you had at the project demonstrations.

Note for you: the sock repo has a hardcoded Wi-Fi password and live patient IDs.



1:35. The blind test.

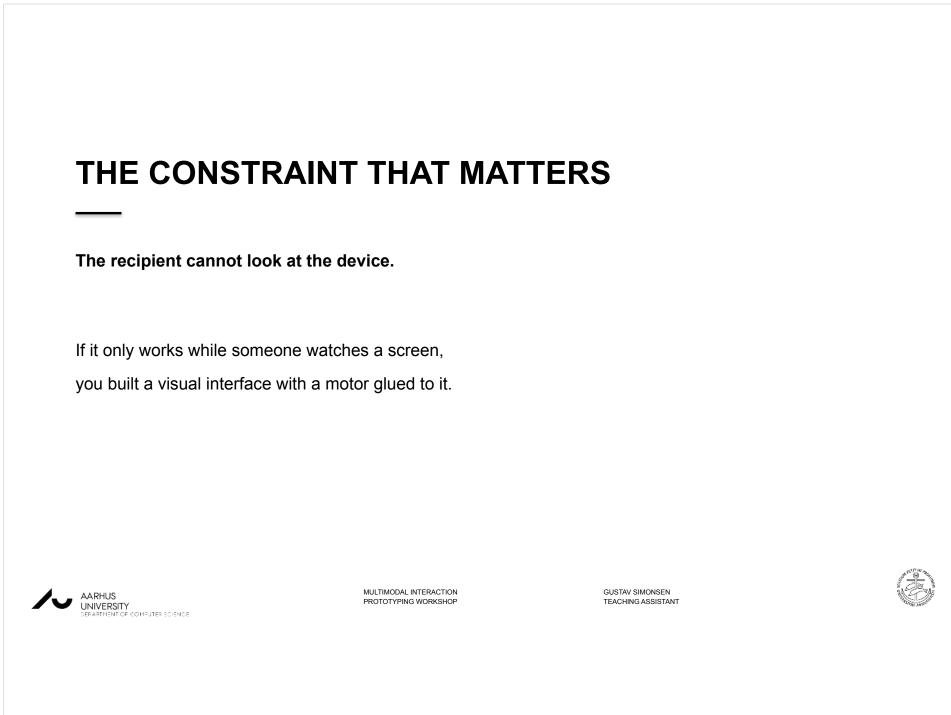
Most groups will already have something from task 05. This is where it gets tested by someone who did not build it.



Read the brief out once, then get out of the way. Full wording, since the slide is only the bones:

"Pick one actuator. Encode three messages: arrived, something wrong, finished. Vary only the rhythm of the pulses and how strong they are. Then hand it to another group with the screen turned away and see if they can name all three."

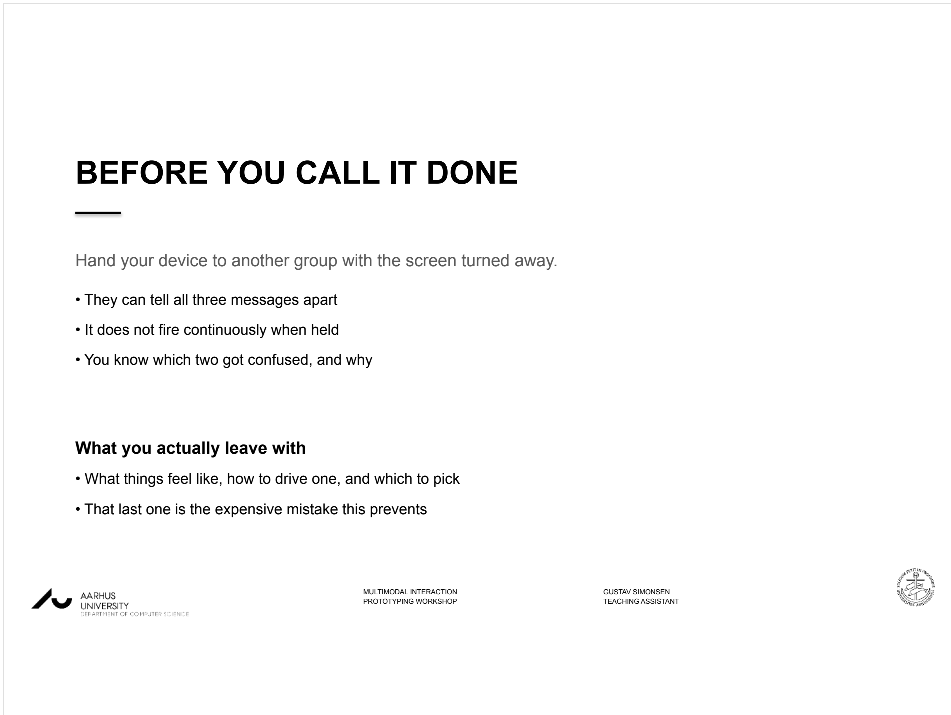
If someone asks to add a second actuator, say no. The constraint is the exercise.



The line worth repeating twice.

Run the test: swap groups, receiver looks away, guess all three.

Then have each group name which two got confused and what would have separated them. That confusion is the actual learning, not the successful ones.



Put this up while the blind tests run, so groups can check themselves
without asking you.

The middle line matters most: a device that fires continuously when held is
the single most common bug, and it is a debounce problem.

Then close on the bottom half, thirty seconds. If they leave with one thing,
make it the last point: pick the actuator before designing the interaction.
Discovering in week 46 that the chosen one cannot do the job is the expensive
mistake this session exists to prevent.

Point them at the repo. Every task sketch is commented for them, and the bench test sketches are there too if they want to see the minimal version.

Last thing to say: pick your actuator before you design the interaction, not after. Today was so they can do that from experience rather than a datasheet.

Pack down: everything back in the box, including the resistors. It will not all come back.

If anyone wants to borrow parts for the project weeks, now is when they ask.